

Adolescent BMI & Behavior Analysis

YRBS 2007 | N = 12,845 | One-Page Summary

01 Research Questions

ANOVA Question

Do weekly exercise levels (Low/Med/High) and biological sex significantly influence adolescent BMI percentiles? Is there a significant interaction effect?

Regression Question

Do daily TV watching hours and weekly exercise days significantly predict BMI percentiles?

02 Data & EDA

Target Y = BMIPCT (0–100)

Predictors (X):

- Sex_Label (Male / Female)
- Activity_Group (Low / Med / High)
- PhysicalActivity5OrMoreDays (0–7 days)
- TelevisionWatching (1–7 ordinal tiers)

64.85
Mean percentile

27.50
SD

↑
Right Skew overweight

03 Methods & Results

One-Sample t-test ($\mu_0 = 50$)

t = 61.19, p < 0.001 → Reject H₀
95% CI: [64.38, 65.32] (Strictly excludes $\mu_0 = 50$)

Two-Way ANOVA (Sex × Activity)

p < 0.001
Sex Effect

p = 0.0239
Activity Effect

p = 0.00035
Interaction ♦

✗ Reject H₀ — Interaction effect confirmed (p = 0.00035)

Multiple Linear Regression (OLS)

R² = 0.012, p < 0.001 → Significant
TelevisionWatching : $\beta = +1.64$, p < 0.001 → Reject H₀
PhysicalActivity5OrMoreDays : $\beta = -0.05$, p = 0.564 → Fail to Reject

04 Conclusions

✗ t-test ($\mu_0 = 50$)
REJECT H₀
t = 61.19, p < 0.001

✗ ANOVA Interaction
REJECT H₀
p = 0.00035

✓ TV Watching β
REJECT H₀
 $\beta = +1.64$, p < 0.001

✗ Exercise Days β
FAIL TO REJECT
 $\beta = -0.05$, p = 0.564

Key Insights

📺 Sedentary Hazard
TV hours are the dominant driver pushing up BMI ($\beta=+1.64$)

⚡ Exercise Paradox
Exercise benefits are non-linear & gender-dependent — not a simple linear effect

🎯 Policy Action
Gender-tailored interventions:
screen-time limits ♂,
optimized fitness environments ♀

Figure 1 : Right-Skewed Distribution of BMI Percentiles

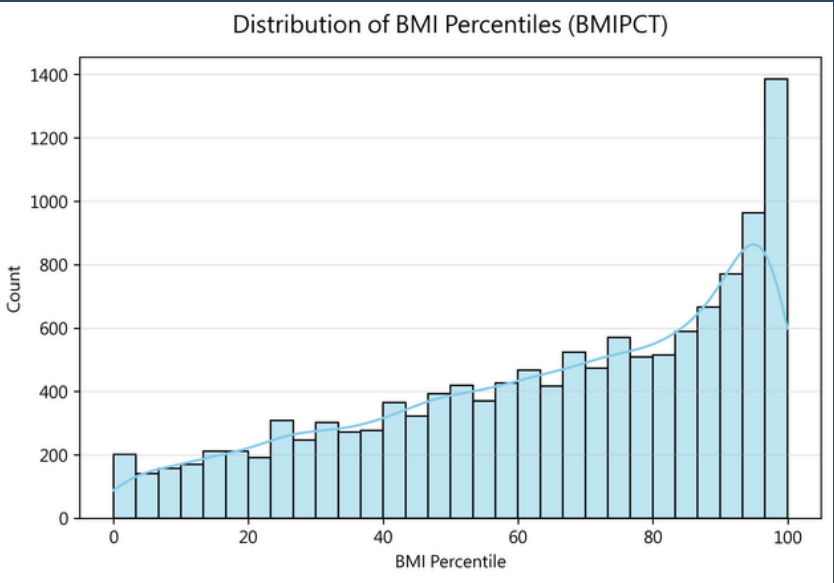


Figure 2 : Two-Way ANOVA Interaction Plot

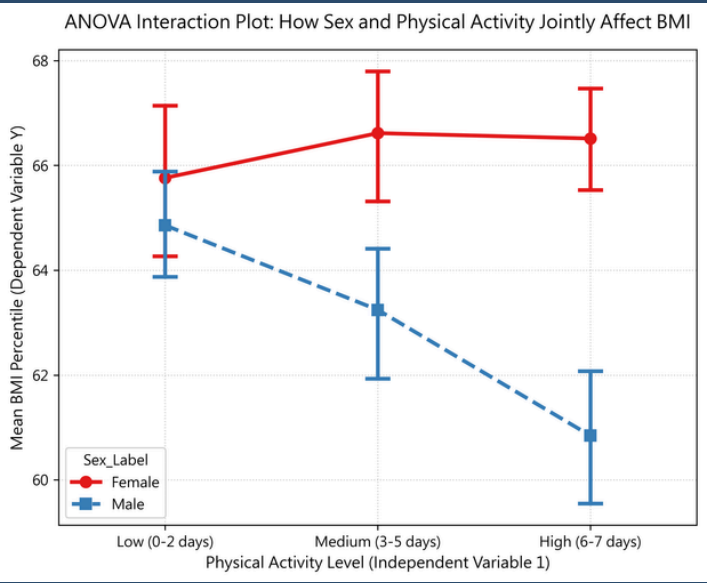


Figure 3 : Regression Coefficient Forest Plot

